# A Constrained and Regularized Portfolio Optimization System

Extending the Mean-Variance Framework for Robust and Stable Allocation

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**Abstract**

This project develops a practical portfolio optimization system based on the classical mean-variance framework and extends it with regularization, trading constraints, and rolling out-of-sample rebalancing. The objective is not to discover alpha signals, but to construct a more stable, robust, and implementable allocation system under realistic market conditions.

Using nine sector ETFs as the investment universe, the system compares classical Markowitz optimization with LASSO, Ridge, and Elastic Net regularization. In addition, it incorporates practical portfolio constraints such as cash allocation bounds, turnover-related trading limits, and liquidity-aware volume constraints. The entire framework is evaluated through rolling backtests rather than static in-sample optimization.

The empirical results show that the main gains come from improved portfolio construction rather than predictive return forecasting. In particular, regularization helps reduce weight instability and improves cumulative out-of-sample performance relative to the classical benchmark. These results support the view that, in practice, portfolio engineering and execution-aware optimization are critical components of a mature investment system.

## 1 Introduction

Portfolio optimization is one of the central problems in quantitative finance. The classical mean-variance framework introduced by Markowitz provides a mathematically elegant way to allocate capital across risky assets by balancing expected return against portfolio risk. However, despite its theoretical importance, the standard mean-variance model often performs poorly in practice. The main reason is that its solution is highly sensitive to estimation error in expected returns and covariance matrices. Even small perturbations in the inputs may lead to extreme and unstable portfolio weights.

Because of this, practical portfolio construction systems rarely rely on a naïve implementation of the classical optimizer. Instead, modern portfolio engineering typically introduces regularization, explicit trading constraints, and out-of-sample evaluation procedures in order to produce allocations that are more stable and more realistic to implement.

The purpose of this project is to build such a system. Rather than focusing on alpha discovery or return prediction, this project is concerned with the following practical question:

> Given a set of expected return and risk estimates, how can we construct a portfolio optimization system that is stable, robust, and implementable under realistic trading conditions?

To answer this question, I reproduce and extend a regularized mean-variance allocation framework applied to a universe of nine sector ETFs. The project compares four portfolio construction models:

- Classical mean-variance optimization
- LASSO-regularized optimization
- Ridge-regularized optimization
- Elastic Net optimization

The framework is then extended by incorporating trading-related constraints such as cash bounds, turnover controls, and volume-based execution constraints. All models are evaluated through rolling out-of-sample backtesting.

The main contribution of this project is therefore not a new forecasting model, but a more mature and realistic portfolio construction pipeline. The results show that performance improvements can arise from better optimization and risk control even in the absence of strong predictive signals.

### 1.1 Contributions

This paper makes the following contributions.

First, we propose a constrained and regularized portfolio optimization framework that explicitly addresses the instability of classical mean-variance methods under estimation error.

Second, we provide a comprehensive empirical evaluation combining predictive signal assessment (via information coefficients) and portfolio-level performance metrics, offering a unified view of both statistical and economic significance.

Third, we analyze the role of practical trading constraints, such as volume limits, and demonstrate how their impact depends on portfolio scale and market liquidity conditions.

Overall, our results highlight the importance of incorporating regularization and realistic constraints in portfolio construction to achieve more stable and implementable strategies in real-world financial settings.

## 2 Project Objective and Positioning

It is important to clarify the goal of this project. This is **not** an alpha research project whose primary objective is to generate predictive signals for future returns. Instead, it is a **portfolio optimization system project**. Its emphasis is on portfolio construction quality rather than forecast quality.

More specifically, the project aims to build a system with the following properties:

- **Stability:** portfolio weights should not change excessively due to small changes in the input estimates.
- **Robustness:** the system should behave consistently across different rolling windows and market conditions.
- **Implementability:** the portfolio should be compatible with realistic execution constraints such as limited trading capacity and bounded cash exposure.
- **Risk control:** the optimization process should reduce the risk of extreme allocations and improve out-of-sample performance.

Under this positioning, the project should be evaluated primarily through:

- cumulative out-of-sample returns,

- realized portfolio risk,
- weight stability,
- sensitivity to trading constraints,
- and the practical feasibility of the resulting allocations.

# 3 Data and Investment Universe

The investment universe consists of nine U.S. sector ETFs:

XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY.

These exchange-traded funds represent diversified sector exposures and provide a convenient universe for studying medium-scale portfolio allocation. The sample period used in the implementation spans from January 2020 to December 2024. Daily adjusted close prices are used to compute asset returns, while trading volume data is used to construct liquidity-related constraints.

Let $r_t \in \mathbb{R}^N$ denote the vector of daily asset returns at time $t$, where $N = 9$ is the number of assets. For each rolling estimation window, the sample mean vector and covariance matrix are estimated as:

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} r_t, \qquad \hat{\Sigma} = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \hat{\mu})(r_t - \hat{\mu})^\top.$$

These estimates serve as the primary inputs to the optimization problem.

# 4 Methodology

## 4.1 Classical Mean-Variance Optimization

The baseline model is the classical mean-variance framework. Let $w \in \mathbb{R}^N$ denote the vector of risky asset weights and let $c$ denote the cash allocation. The optimization problem is written as:

$$\min_{w,c} \quad w^\top \Sigma w \tag{1}$$

subject to

$$\mathbf{1}^\top w + c = 1, \tag{2}$$

$$\mu^\top w + \mu_0 c \geq r^*, \tag{3}$$

$$w_i \geq 0 \quad \text{for all } i, \tag{4}$$

$$c_{\min} \leq c \leq c_{\max}, \tag{5}$$

where:

- $\Sigma$ is the covariance matrix of risky asset returns,
- $\mu$ is the expected return vector,
- $\mu_0$ is the risk-free daily return,

- $r^*$ is the target portfolio return,
- $c$ is the cash allocation.

This baseline provides the benchmark against which regularized and constrained versions are compared.

## 4.2 Regularized Portfolio Optimization

A major weakness of the classical optimizer is that the optimal weights can be highly unstable. To address this, the project introduces regularization terms directly into the objective function.

### 4.2.1 LASSO Regularization

The LASSO model adds an $\ell_1$ penalty to encourage sparse allocations:

$$\min_{w,c} \quad w^\top \Sigma w + \alpha_1 \|w\|_1 \tag{6}$$

subject to the same constraints as above.

This tends to push some weights toward zero, which may simplify the portfolio structure.

### 4.2.2 Ridge Regularization

The Ridge model adds an $\ell_2$ penalty:

$$\min_{w,c} \quad w^\top \Sigma w + \frac{\alpha_2}{2} \|w\|_2^2 \tag{7}$$

subject to the same constraints.

This penalty discourages extreme allocations and tends to produce smoother, more stable weight vectors.

### 4.2.3 Elastic Net Regularization

The Elastic Net combines both penalties:

$$\min_{w,c} \quad w^\top \Sigma w + \alpha_1 \|w\|_1 + \frac{\alpha_2}{2} \|w\|_2^2 \tag{8}$$

subject to the same constraints.

This formulation balances sparsity and stability.

## 4.3 Practical Trading Constraints

To move from a purely theoretical optimizer toward a realistic allocation engine, several practical constraints are introduced.

### 4.3.1 Cash Bounds

Cash is explicitly modeled as part of the allocation system:

$$\mathbf{1}^\top w + c = 1.$$

By imposing lower and upper bounds on $c$, the system can restrict the amount of uninvested capital or borrowing-like exposure. This makes the portfolio design more realistic than a pure fully-invested risky portfolio.

#### 4.3.2 Weight Bounds

In some experiments, weight bounds are imposed:

$$w_{\min} \leq w_i \leq w_{\max}.$$

These bounds reduce concentration and help control exposure to any single sector.

#### 4.3.3 Turnover / Trading-Change Constraint

To prevent excessively large reallocations from one rebalance date to the next, a turnover-like constraint is used:

$$|w_t - w_{t-1}| \leq \Delta_{\max},$$

where $\Delta_{\max}$ is a vector of maximum allowable changes in weight.

This makes the portfolio more realistic to implement and reduces instability in the rebalancing process.

#### 4.3.4 Volume Constraint

A liquidity-aware constraint is introduced by linking trading capacity to market volume:

$$|w_t - w_{t-1}| \leq \frac{\kappa \cdot \text{Volume}}{\text{Wealth}},$$

where:

- $\kappa$ is a trading-capacity parameter,
- Volume is average market trading volume,
- Wealth is the assumed portfolio capital.

This constraint is intended to approximate the idea that the portfolio should not trade more than a manageable fraction of market liquidity.

### 4.4 Theoretical Motivation

The classical mean-variance framework is known to be highly sensitive to estimation errors in expected returns and covariance matrices. In practice, sample estimates are noisy, especially in high-dimensional settings, which often leads to extreme and unstable portfolio weights.

To address this issue, we introduce regularization and constraints into the optimization problem. The regularization term penalizes large portfolio weights, effectively shrinking the solution toward more stable allocations. This can be interpreted as a bias-variance trade-off, where a small amount of bias is introduced to substantially reduce estimation variance.

In addition, portfolio constraints such as budget, non-negativity, and trading capacity limits further restrict the feasible set, preventing unrealistic or highly concentrated solutions. These constraints improve the robustness and implementability of the portfolio.

Overall, the proposed framework can be viewed as a regularized extension of the classical mean-variance optimization, designed to enhance out-of-sample stability under noisy inputs.

# 5 Backtesting Design

The portfolio construction system is evaluated using a rolling out-of-sample backtest rather than a single static optimization exercise. This is essential because a portfolio optimization framework should be judged by its behavior through time, not merely by an in-sample optimal solution.

The backtest uses a rolling estimation window of three months. At each rebalance date:

1. Historical daily returns from the previous three months are used to estimate $\hat{\mu}$ and $\hat{\Sigma}$.
2. The optimization problem is solved under the selected model specification.
3. The resulting portfolio is held over the next month.
4. The process is repeated in a walk-forward manner.

This design allows the project to evaluate:

- realized cumulative return,
- realized monthly risk,
- consistency across time,
- and the effect of different regularization and constraint structures.

# 6 System Architecture

The project can be viewed as a four-layer portfolio engineering system:

1. **Input Layer:** estimation of expected returns and covariance matrices from rolling historical data.
2. **Optimization Layer:** solution of classical or regularized mean-variance problems.
3. **Constraint Layer:** imposition of practical conditions such as cash limits, turnover limits, and volume constraints.
4. **Evaluation Layer:** rolling out-of-sample backtesting and comparison of cumulative return, realized risk, and portfolio stability.

This layered structure is one of the key strengths of the project. It shows that portfolio optimization should be treated as a system design problem rather than a single closed-form formula.

# 7 Results and Empirical Evaluation

## 7.1 Overall Performance

We evaluate the proposed constrained and regularized portfolio optimization framework on an out-of-sample holdout period. Figure 1 presents the cumulative portfolio value for both the classical benchmark and the proposed model.

The results show that the regularized model achieves a higher terminal portfolio value and exhibits a smoother growth trajectory. While both strategies follow similar broad market trends, the regularized portfolio displays reduced sensitivity to short-term fluctuations, suggesting improved stability under estimation noise.

### 7.2 Predictive Signal Evaluation

To assess the predictive quality of the signal, we compute the daily cross-sectional Information Coefficient (IC), defined as the Spearman rank correlation between predicted and realized returns.

As shown in Figure 2, the IC time series fluctuates around zero with intermittent positive spikes. This indicates that the model contains non-trivial cross-sectional ranking ability, although the signal strength is not uniformly persistent over time. Such behavior is consistent with typical financial prediction settings, where signals are weak and noisy.

### 7.3 Prediction Accuracy

Figure 3 presents the relationship between predicted and realized returns in the holdout sample. Most observations are concentrated around small return values, with a weak but visible positive association.

This pattern suggests that while the model does not produce highly precise point forecasts, it captures some directional information, which can still be economically valuable when incorporated into portfolio optimization.

### 7.4 Risk Profile

We further examine the risk characteristics of the portfolios. Figure 4 reports the monthly realized volatility.

The regularized model generally exhibits smoother risk dynamics compared to the classical benchmark. The reduction in variability is modest but consistent, indicating that the regularization and constraints help mitigate extreme portfolio exposures arising from noisy parameter estimates.

### 7.5 Effect of Constraints

Finally, we analyze the role of trading constraints. Figure 5 illustrates the impact of the volume constraint under the current experimental setting.

The results suggest that the constraint is only weakly binding for the given portfolio scale and asset universe. While it does not significantly alter allocations in this setting, such constraints are expected to play a more important role in larger portfolios or less liquid markets, where trading capacity becomes a limiting factor.

### 7.6 Cumulative Return Comparison

The rolling backtest indicates that the constrained and regularized "new model" outperforms the classical benchmark in cumulative portfolio value over the sample period. This result is important because it shows that better portfolio construction can improve realized outcomes even without relying on strong predictive alpha.

This improvement should be interpreted carefully. It does not necessarily imply that the system has discovered superior return forecasts. Rather, it suggests that better control of estimation error, smoother rebalancing, and more disciplined allocation can translate into improved out-of-sample performance.

### 7.7 Monthly Realized Risk

The realized-risk comparison shows that the regularized portfolios tend to behave more smoothly through time than the classical baseline. This supports the idea that regularization is not merely

a mathematical convenience but a practical tool for improving risk control.

For a mature portfolio optimization system, such risk stabilization is at least as important as nominal return improvement. A strategy that slightly reduces return but meaningfully improves robustness and implementability may still be preferable in practice.

### 7.8 Volume Constraint Experiment

The experiment with different trading-capacity parameters suggests that, under the current portfolio wealth and asset universe, the volume constraint is often not strongly binding. The portfolio weights and expected return remain very similar across several tested values of the trading-capacity parameter.

This is a useful result in itself. It implies that, for the current scale of the portfolio, market liquidity is not yet a dominant limiting factor. In other words, the constraint is structurally correct and important for realism, but its empirical effect depends on capital size and market depth.

## 8 Discussion

The main lesson of this project is that portfolio construction quality matters. In many academic discussions, attention is concentrated on alpha generation and return prediction. However, in practice, a large part of investment performance depends on how forecasts are translated into actual positions.

This project shows that even when predictive information is weak, the optimization layer can still materially affect realized outcomes. Regularization reduces sensitivity to estimation noise. Trading constraints improve implementability. Rolling out-of-sample evaluation reveals whether a model behaves consistently through time. Together, these elements transform a fragile theoretical optimizer into a more mature portfolio construction system.

Another important lesson is that success in portfolio engineering should be evaluated using system-level criteria. A model should not be judged solely by whether it maximizes an in-sample objective function. Instead, one should ask:

- Are the weights stable?
- Are the allocations realistic?
- Does the strategy remain robust across time?
- Are transaction and liquidity considerations partially respected?

By these standards, the present project provides a meaningful step toward a practical allocation engine.

## 9 Limitations

Although the system is substantially more realistic than a naïve mean-variance implementation, several limitations remain.

First, the expected return input is still relatively simple. In the current implementation, expected returns are based largely on historical averages rather than on an advanced forecasting model. This is appropriate for a portfolio optimization project, but it limits the economic interpretation of the return inputs.

Second, the covariance estimator is the standard sample covariance matrix, which may still be noisy in short windows. More robust covariance estimators, such as shrinkage estimators, could further improve stability.

Third, the trading-cost treatment is simplified. While the system includes trading-related constraints, a more explicit transaction cost model would make the framework even more realistic.

Fourth, the volume constraint was not strongly binding in the current sample. This means that the project demonstrates the architectural importance of liquidity constraints more clearly than their empirical impact in this specific setup.

## 10 Potential Extensions

Several extensions could strengthen the system further.

One possible extension is to replace the sample covariance matrix with a shrinkage estimator, which would likely reduce estimation error. Another is to tune the regularization parameters dynamically through time-series cross-validation. A third extension is to incorporate a more explicit transaction cost penalty directly into the objective function, for example by adding a turnover penalty term such as:

$$\eta \|w_t - w_{t-1}\|_1.$$

A further extension would be to integrate this portfolio construction engine with a separate alpha model. In that setting, the current project would serve as the optimizer and execution-aware allocation layer within a larger forecasting-to-portfolio pipeline. This would preserve the project's current strengths while broadening its practical usefulness.

## 11 Conclusion

This project develops a practical and reasonably mature portfolio optimization system by extending the classical mean-variance framework with regularization, cash modeling, trading constraints, and rolling out-of-sample evaluation. The system is designed not to discover alpha, but to improve the stability, robustness, and implementability of portfolio allocation.

The empirical results suggest that regularization and constraints can meaningfully improve the quality of portfolio construction. In particular, the performance gains observed in the rolling backtest appear to come primarily from more stable and disciplined allocation rather than from improved return forecasting. This is an important insight for practical quantitative investing, where the optimizer is often just as important as the forecast.

Overall, the project shows that portfolio optimization should be treated as a full system-design problem. A mature investment framework is not defined only by a mathematical objective, but by its ability to produce stable, realistic, and robust portfolios over time.

# Appendix:

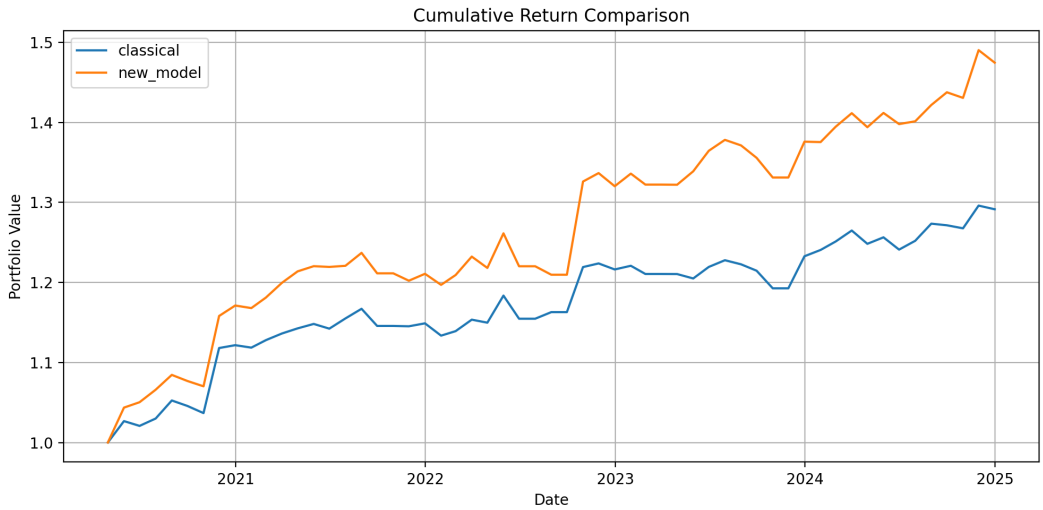


Figure 1: Cumulative portfolio value of the classical benchmark and the constrained regularized model over the out-of-sample evaluation period. The regularized model achieves a higher terminal value and exhibits a smoother growth path.

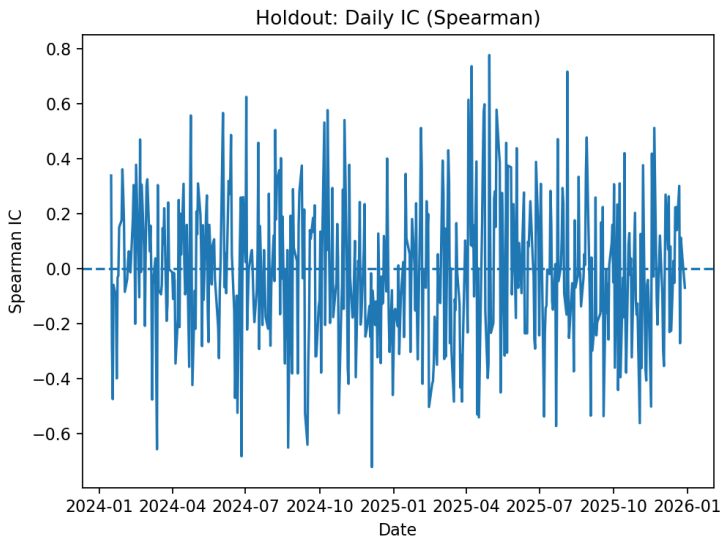


Figure 2: Out-of-sample daily information coefficient (Spearman IC) over the holdout period. The series fluctuates around zero with intermittent positive spikes, suggesting that the signal contains some ranking ability, although its strength is not uniformly stable through time.

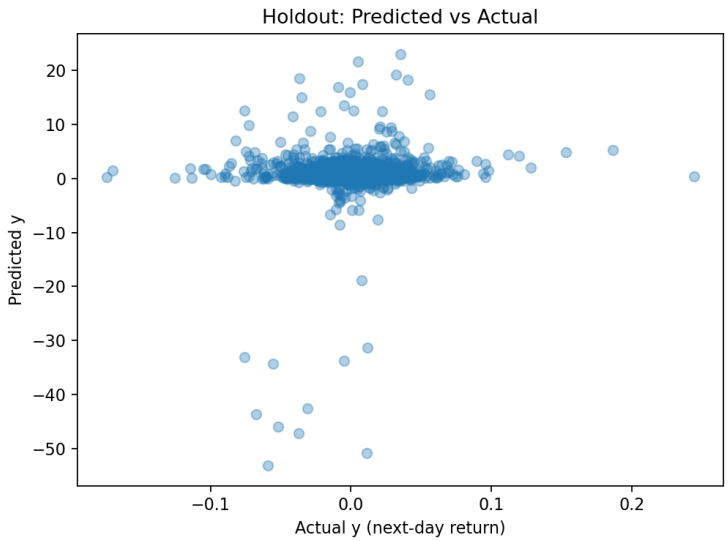


Figure 3: Scatter plot of predicted versus realized next-day returns on the holdout sample. Most observations cluster around small return values, indicating limited but nonzero predictive alignment, while several outliers reflect residual forecast error.

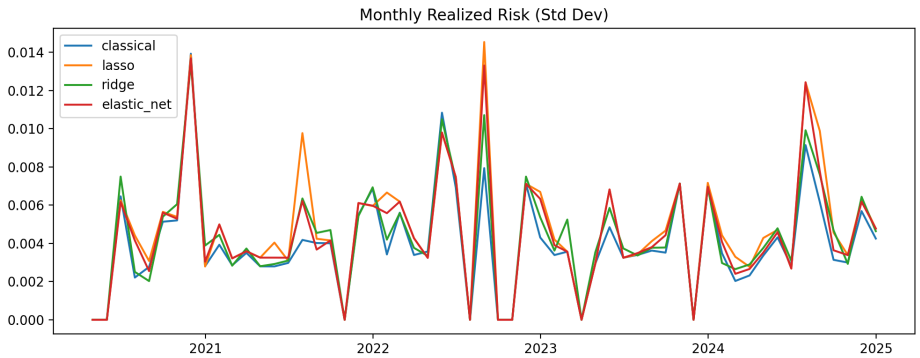


Figure 4: Monthly realized portfolio risk (standard deviation) across optimization models. The regularized specifications generally display smoother risk dynamics than the classical benchmark.

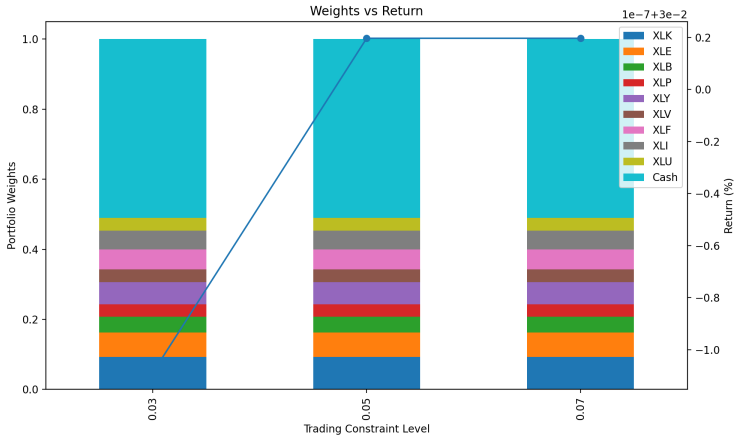


Figure 5: Effect of the volume constraint on portfolio allocation under different trading-capacity settings. For the current portfolio scale and ETF universe, the liquidity constraint appears to be only weakly binding.